

4. Flight states overview

This chapter provides an overview of various states of a system flight record (SFR), giving the necessary context for many of the system's functions.

4.1. Lifecycle State

Lifecycle states represent the states of an SFR within the system. The following lifecycle states are defined:

Lifecycle State	Description
Proposed	An SFR is created upon successful processing of an ICAO FPL or ADEXP IFPL message. The SFR is ready to be handled by the system. If not activated, after a configurable timeout the SFR will progress to Terminated state.
Active	When an update allowing for correlation is received (a notification, an estimate, a departure is reported or detected) the SFR progresses to Active state. Only an Active flight can be taken under control (i.e. assumed).
Terminated	After a flight is decontrolled by the last sector within the AoR or an arrival is reported or detected, the SFR progresses to Terminated state. After a configurable timeout a Terminated SFR is removed from the system. A Terminated SFR can be returned to an Active state.

4.2. Control State

Control state, as defined in the table below, is maintained per Controlling Sector, for each Active SFR.

Control State	Meaning
Non-Concerned	Flight does not enter a Controlling Sector and the Controlling Sector has not assumed control of the flight.
Concerned	Flight will enter a Controlling Sector, but the Controlling Sector has not assumed the flight yet.
Transfer-In Initiated	The upstream Controlling Sector (or upstream centre - in future release) has initiated transfer-of-control to the Controlling Sector. The Controlling Sector has not assumed control yet.
Request-In Initiated	The Controlling Sector has performed a ROF (Request on Frequency) operation for an SFR, while still in coordination phase.
Assumed	The Controlling Sector has assumed control of the flight.

Control State	Meaning
Transfer-Out Initiated	The Controlling Sector has initiated transfer-of-control to the next downstream Controlling Sector (or downstream centre - in future release). Downstream Controlling Sector (or centre) has not assumed control yet.
Request-Out Initiated	The upstream Controlling Sector (or upstream centre - in future release) has instructed the flight to change frequency and preforms COF (Change of Frequency) operation.
Redundant	The downstream Controlling Sector (or downstream centre - in future release) has assumed control of the flight, but the flight is still positioned in the Controlling Sector's volume.
Completed	The flight is no longer under control of the given Controlling Sector and is not positioned within its volume, and the flight does not plan to enter the Controlling Sector again.

4.3. Coordination State

The state assigned to the flight which a particular SFR represents, based on each particular coordination dialog. This means that the SFR can have multiple Coordination States, depending on the number of coordination dialogs.

Inbound Coordination States:

Coordination State	Meaning
None	State set to None when a MAC message is received.
Notified	State set to Notified when an ABI message is received.
Coordinated	State set to Coordinated when an ACT message is received or the flight is manually activated by providing an estimate (following voice communication with upstream centre controller).

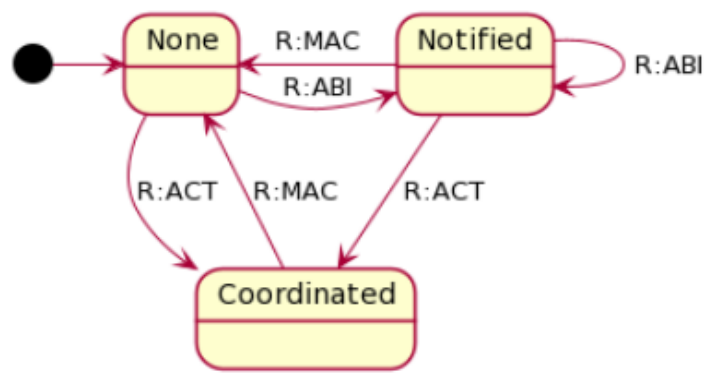


Figure 15. Inbound Coordination State Transition Diagram

Outbound Coordination States:

Coordination State	Meaning
None	State set to None when a MAC message has been acknowledged with a received LAM or a manual action, and the flight is not on route to enter the coordinating FIR.
Notifying	State set to Notifying after an ABI (or Revised ABI) has been sent.
Notified	State set to Notified after an ABI (or Revised ABI) has been acknowledged with a received LAM or a manual action.
Coordinating	State set to Coordinating after an ACT has been sent.
Coordinated	State set to Coordinated after an ACT has been acknowledged with a received LAM or a manual action.
Revising	State set to Revising when coordinated transfer conditions are revised, through revision, but not yet acknowledged with a received LAM.
Abrogating	State set to Abrogating after a MAC has been sent.

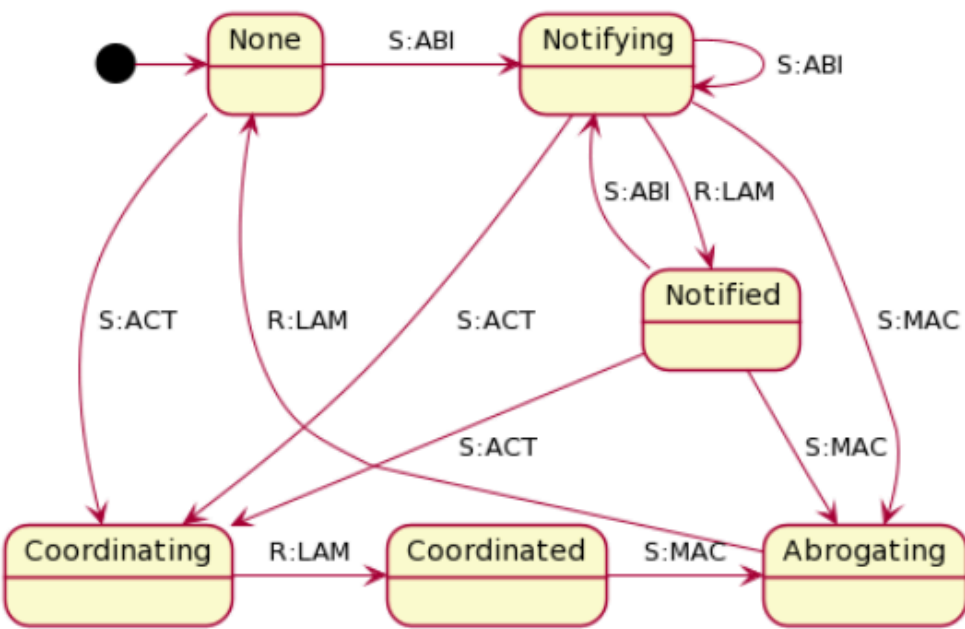


Figure 16. Outbound Coordination State Transition Diagram

4.4. Track Representation of Flight States

The control and coordination states of a flight are shown as colour changes to the track label fields, leader line, track symbol. If enabled, the history dots and speed vector also change colour. Flights will also reflect changed field colouring in flight lists. The sector indicator field, one to the right of the callsign, generally shows the control sector in control of the flight but can take different values to show intermediary states.

NOTE

The leader line, track symbol, history dots and speed vector always have the same color as the callsign field.

Few different flight states are shown in grey indicating no relevance to the controller. They are:

- Non-Concerned flights
- Concerned flights, if no inbound coordination has occurred yet
- Terminated flights



Figure 17. Track Non-Concerned / Concerned With No Coordination / Terminated

When an ABI is received for a Concerned track indicating the the Inter-sector Coordination state is Notified , the callsign is shown in white.

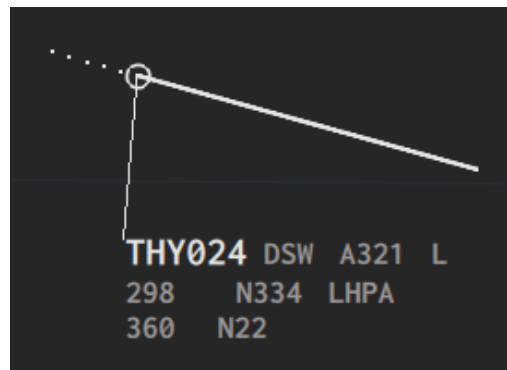


Figure 18. Track Inbound Notified

When an ACT is received for a Concerned track indicating the the Inter-sector Coordination state is Coordinated , all fields are shown in white.



Figure 19. Track Inbound Coordinated

When the Control State of an Inbound Coordinated track is set to Transfer-In Initiated , the callsign is shown in blue. Indicating that the track can be Assumed.

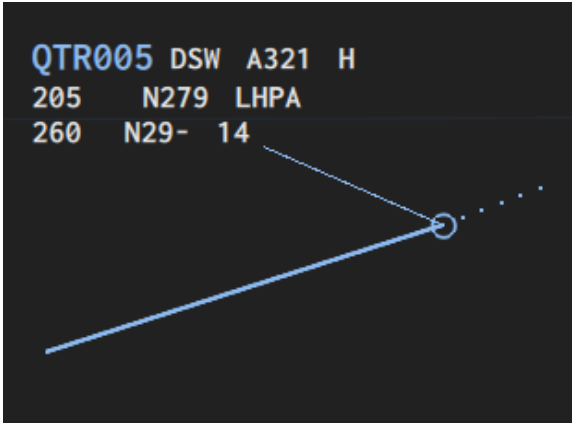


Figure 20. Track Transfer-In Initiated

When the Control State of a track is set to 'Request-In Initiated', ROF is shown beside the callsign to indicate the aircraft has been requested on this frequency.

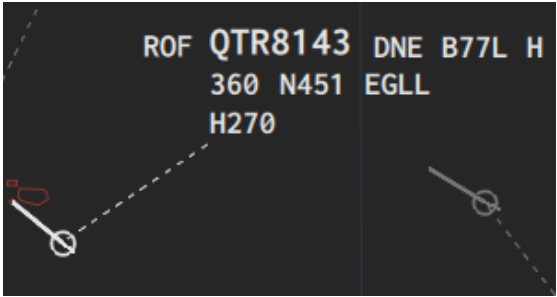


Figure 21. Track Request-In Initiated

When the Control State is Assumed and the Outbound Coordination has not occurred yet, all fields are shown in blue.

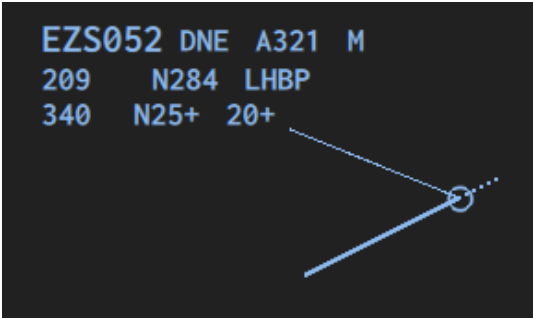


Figure 22. Track Assumed

When an ACT is sent for an Assumed track indicating the Outbound Coordination state is Coordinating and no answer has been received causing the coordination to time out. In this case, the sector indicator shows the next sector and is colored grey, which indicates that the outbound coordination has not been successful yet. A phone icon is shown beside the callsign to indicate the time out, the icon can be clicked to open a manual coordination dialog.

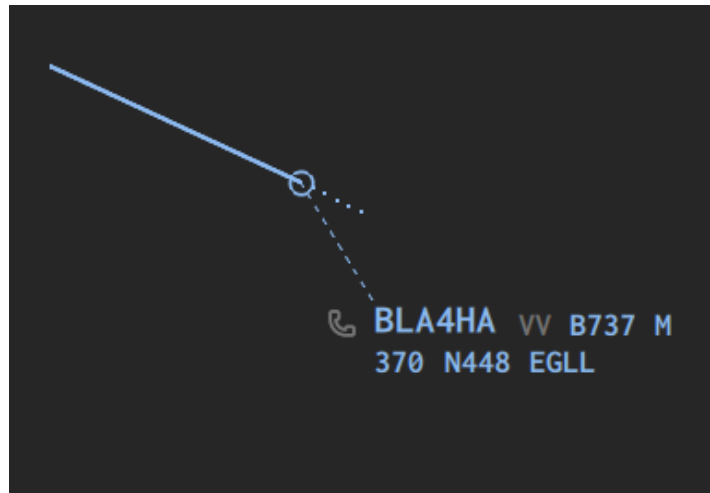


Figure 23. Track Outbound Coordinating and timed out

When a LAM is received for an Assumed track indicating the Outbound Coordination state is Coordinated . The sector indicator shows the next sector and is colored white. This indicates that the track is ready to initiate transfer out.

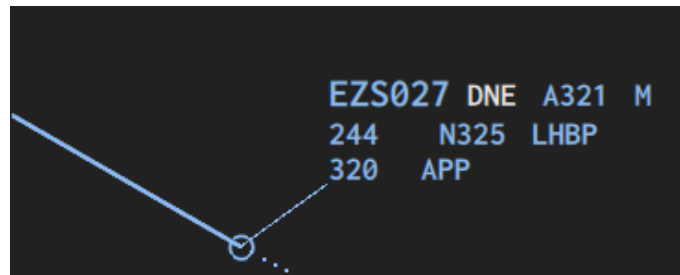


Figure 24. Track Outbound Coordinated

When the Control State is Transfer-Out Initiated , the callsign is colored copper, the sector indicator shows the next sector and is colored white.

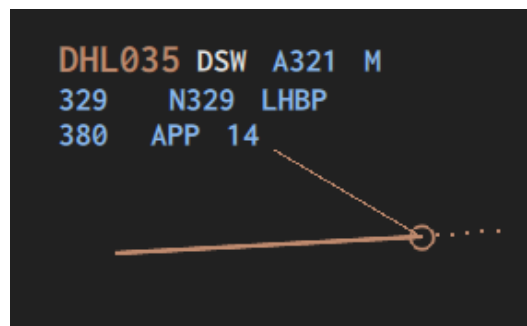


Figure 25. Track Transfer-Out Initiated

When the Control State is Request-Out Initiated , ROF is shown beside the callsign to indicate the aircraft has been requested on another frequency. The sector indicator shows the next sector and is colored white.

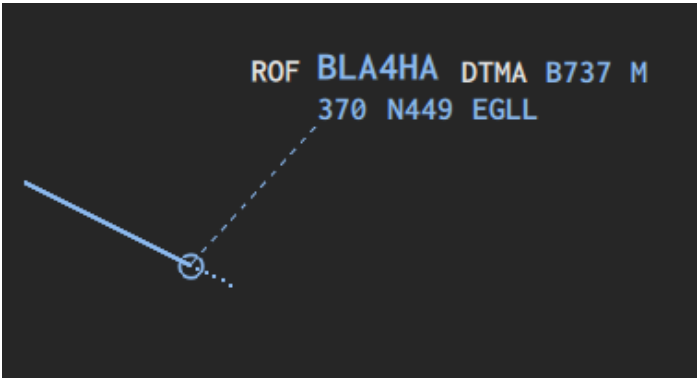


Figure 26. Track Request-Out Initiated

When the Control State is Redundant , all fields are shown in copper.

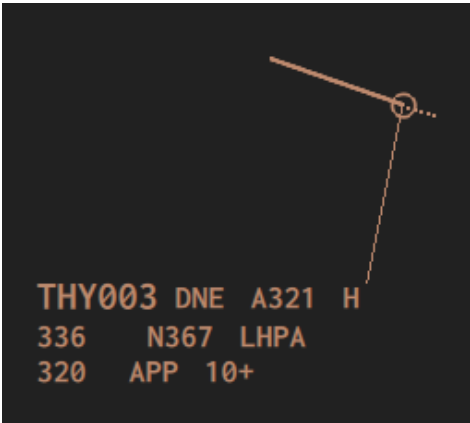
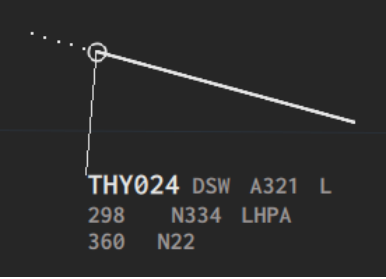
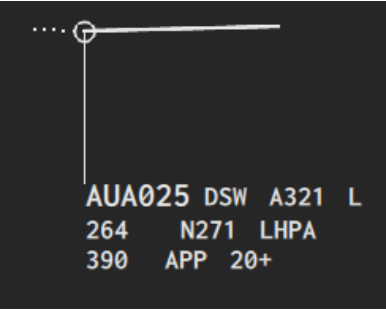
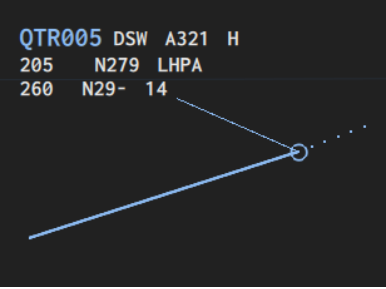
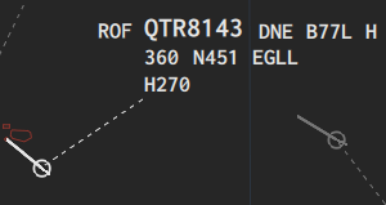
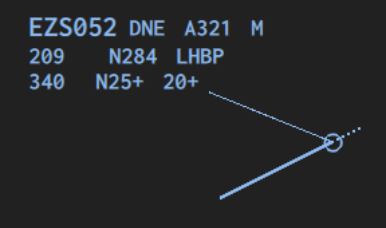
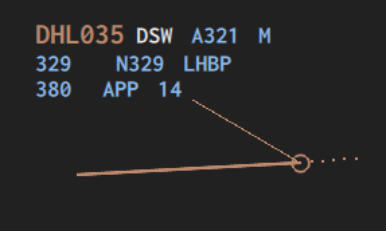
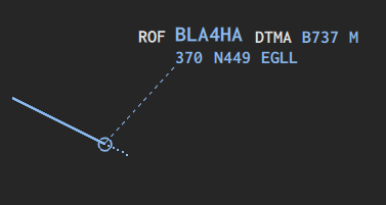
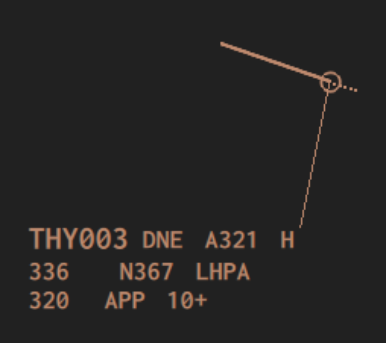
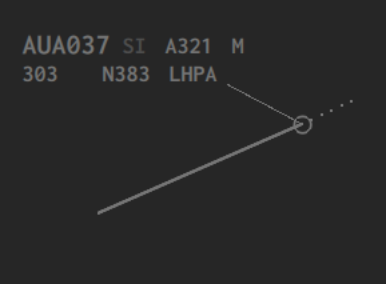


Figure 27. Track Redundant

The following table lists the same information specified above in a format that might be easier to quickly reference.

Track Label	Control State	Inbound Coordination	Outbound Coordination	Sector Indicator	Other
	Non-Concerned	N/A	N/A	Controlling Sector	
	Concerned	None	N/A	Controlling Sector	

Track Label	Control State	Inbound Coordination	Outbound Coordination	Sector Indicator	Other
	Concerned	Notified	N/A	Controlling Sector	CWP controlling sector is the next sector for the track
	Concerned	Coordinated	N/A	Controlling Sector	CWP controlling sector is the next sector for the track
	Transfer-In Initiated	Coordinated	N/A	Controlling Sector	
	Request-In Initiated	Coordinated	N/A	Controlling Sector	
	Assumed	N/A	None, Notifying, Notified, Coordinating, Abrogating	Controlling Sector	
	Assumed	N/A	Coordinating	Next Sector	Outbound coordination timed out

Track Label	Control State	Inbound Coordination	Outbound Coordination	Sector Indicator	Other
	Assumed	N/A	Coordinated	Next Sector	
	Transfer-Out Initiated	N/A	Coordinated	Next Sector	
	Request-Out Initiated	N/A	Coordinated	Next Sector	
	Redundant	N/A	N/A	Controlling Sector	
	Completed	N/A	N/A	Controlling Sector	